HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Assisted in defining and implementing **quality standards** and specifications for medical device manufacturing processes, ensuring adherence to **corporate and regulatory requirements**.
- Applied knowledge of **Six Sigma** and **statistical analysis** to investigate production exceptions, developing data-driven solutions that improved process efficiency by approximately **15%**.
- Collaborated effectively with cross-functional team members including R&D;, Quality Assurance, and Operations to support day-to-day operating requirements and troubleshoot production issues.
- Participated in new product and technology transfers, gaining exposure to the design and development lifecycle of next-generation medical technologies.
- Supported day-to-day operating requirements by assisting in the analysis of **customer feedback and complaints**, contributing to product improvement initiatives.
- Demonstrated proactive support for environmental management requirements by adhering to site protocols and participating in relevant team discussions.
- Containerized Java Spring Boot applications using **Docker** and deployed them to an **AWS EC2** environment, managing CI/CD pipelines via **Jenkins**.

SKILLS

Manufacturing Engineering – Quality Standards, Six Sigma, Lean Principles, Process Improvement, New Product Introduction, Validation Techniques, Regulatory Requirements, Environmental Management

Programming Languages – Java, Python, SQL, R

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn

Databases – PostgreSQL, MySQL, Oracle

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Analytics & BI – Power BI, Tableau, DAX, Power Query, Statistical Analysis

PROJECTS

Manufacturing Process Optimization (HCL Technologies)

- Analyzed production data using **Python** and **pandas** to identify bottlenecks in a simulated manufacturing workflow, pinpointing areas for improvement.
- Developed automated reporting scripts in **Python** to track key performance indicators related to production throughput and defect rates, reducing manual reporting time by approximately **40%**.
- Applied **statistical analysis** techniques to evaluate the impact of process changes, providing data-backed recommendations for quality enhancements.

Quality Dashboard Development (HCL Technologies)

- Designed and implemented interactive **Power BI** dashboards to visualize production quality metrics and operational data for cross-functional teams.
- Developed **DAX measures** and performed data modeling in Power BI to create insightful reports on defect trends and process compliance.
- Integrated data from multiple sources using **Power Query**, ensuring data accuracy and consistency for stakeholder review.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)